



Agri–food supply chain design for perishable products: application to small-scale farmers

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Abstract

Modelling the deterioration of perishable products in logistics systems is quite challenging as it depends on the nature of the product and environmental factors such as humidity and temperature in transport and storage processes. This paper proposes a mixed-integer linear programming model for designing a supply chain network of perishable agricultural products within a strategic and tactical-operational decision framework. The proposed framework considers price sensitivity to product quality, which decreases as the product approaches shelf life, and the possibility of choosing between refrigerated and non-refrigerated facilities, as the type of facility influences the quality and, consequently, the price of the product over the storage period. Based on deterministic scenarios, the model optimises warehouse type selection and location, market assignment to located warehouses, farmer selection and related product purchasing/producing, and inventory evolution, considering warehouse setting, purchasing, inventory, and transport costs, along with income from product selling in the existing markets. The final selling prices in the markets depend on product quality or deterioration level, which relies on how long products are stored in the warehouses. The model is applied to a real-world case of small farmers in Colombia, considering different product demand profiles. Results show that including deterioration of perishability products in supply chain design may significantly impact strategic decisions on network design, estimations of chain revenue, and inventory/capacity relationships. Also, the synchrony between supply and demand strongly influences the network structure. Finally, the choice of refrigerated facilities might effectively increase the profit of the agri-food supply chain over regular facilities.

Keyword Supply chain network design; Perishable products; Inventory-location; Agricultural supply chain

1 Introduction

Supply chains integrate networks of facilities and transport options for the supply, manufacturing, storage, and distribution of materials and products (Garcia and You 2015). Chopra and Meindl (2016) refer to supply chain design as a strategic process that encompasses decisions regarding the number and location of production and storage facilities, the quantity and capacity at each facility, the allocation of activities to resources and supplier selection, the assignment of customers to one or more locations, the choice of transport modes, and the type of information system to use. Its structure or topology has a significant influence on how companies execute strategies, perform, and gain competitive advantages (Calleja et al. 2018).

The design of a supply network is of paramount importance to attaining the objectives of businesses and fulfilling the requirements of multiple products across the globe. If the process is not executed effectively, the implications may be severe. For example, suboptimal choices in the location of facilities can have a significant impact on the flows of the entire supply chain and impede processes such as the distribution of assets. Conversely, effective management of inventory levels and policies can influence customer satisfaction and service through the availability of products and prompt delivery times (Glock et al. 2014). Thus, the process of designing a supply chain network is incredibly intricate, as it involves making numerous decisions within a tight timeframe, all while grappling with factors unique to each modelling context.

Agricultural-supply chains (ASC) are more complex to handle than typical supply chains due to unique, well-reported characteristics, such as price volatility, reduced shelf life (Amorim et al. 2012), demand variability (Ahumada and Villalobos 2009), and dependence on climatic conditions (Salin 1998). Every year, about 1.3 billion tons of agricultural products, such as fruits and vegetables, are wasted worldwide (Gustavsson et al. 2011) due to deficiencies in different stages of the (ASC) (Lemaire and Limbourg 2019). Each supply chain echelon (i.e. production, transport, and storage) contributes to this process (Priefer et al. 2016). In Colombia, there is a worrying situation of inefficiency in the food chains, with 34% of the available quantity of food being lost and wasted annually, i.e., 9.74 million tons (Departamento Nacional de Planeación 2016), with loss indicators (64%) above the world average (54%). This report states that 40.5% of the wasted food (3.95 million tons) is lost in the agricultural production stage and 19.8% (1.93 million tons) in the post-harvest and storage processes. Food losses serve as an indicator of social inequity, as the wasted products could potentially be utilized to nourish deprived populations, including homeless individuals and those with low incomes (Mallidis et al. 2022).

Unfortunately, food losses are present in each stage of the food supply chain (FSC) (i.e. harvest, processing, storage, transport). These losses are stressed by factors such as an inadequate logistics infrastructure (which is prevalent in developing countries) (Kayikci et al., 2022) and the characteristics of the product, whether it is *fresh* or *processed* (van der Vorst 2000). The management of FSC is different from other kinds of supply chains because of its particular features (Amorim et al. 2012), especially those related to agricultural products.

It is crucial to consider the time frame by which a product preserves the (initial) properties or characteristics that make it valuable and desirable to purchase before becoming of no value (Shafiee et al. 2022). Therefore, quality and shelf life are related (Rong et al. 2011). As another stressor of supply chain complexity, the products' quality and freshness significantly impact the customers' purchasing decisions (Mirzaei and Seifi 2015; Rohmer et al. 2019; Lu et al. 2020).

According to Ahumada and Villalobos (2009), there are two types of ASC: perishable and non-perishable products. Perishable products such as fruits and vegetables must be kept refrigerated or frozen throughout the supply chain since temperature is the environmental factor that affects shelf life the most (Wu et al. 2019). If storage temperature is not appropriate, food experiences several chemical and biological reactions (Baloch et al. 2011; Singh et al. 2013), affecting firmness and texture (Baloch and Bibi 2012) and accelerating the ripening process, thereby diminishing product quality and even making it inedible (Shen et al. 2011; Mercier et al. 2017). The loss of quality reduces the price of the product.

The particularities of ASC for perishable products (after this ASCP) add complexity to logistics. Quality changes along the transport and storage stages must be handled in supplying perishable products to consumers (Rong et al. 2011; Yu and Nagurney 2013; Afshar et al. 2022). In addition, holding an adequate inventory level is a critical topic in ASCP management. Low inventory levels may cause economic losses due to unsatisfied demand. Meanwhile, excessive inventory levels represent a higher financial burden (because of holding costs) and potential costs related to wasted products (Chintapalli 2014; Galal and El-Kilany 2016). Therefore, properly designing and managing perishable goods supply chains represent a logistical challenge.

In this paper, we aim to design an ASCP in a real-world case. We found a valuable motivator for this paper by identifying from the literature review that the implications of the type of facilities to locate in the supply chain design process over the quality of agricultural products have not been addressed yet. **Thus, we pose a key research question: how does the type of facility (whether refrigerated or not) affect the design of an ASCP?** Another concern we aimed to tackle with this paper was the impact of such type of facility–location decisions on the revenue perceived from product sales. It is well known that the deterioration of agricultural–food products is highly complex and is often represented by non-linear expressions such as the Weibull power law (Wu 2001; Giri et al. 2003; Skouri et al. 2009; Pal et al. 2014; Chakraborty et al. 2018; Rapolu and Kandpal 2020), initially proposed by Peleg et al. (2002). However, such expressions implicate computationally cumbersome problems. Complexity increases even more when addressing multidimensionality (multiple periods, multiple products). This article intended to model the perishability of agricultural products, considering product depreciation through a practical approach that reflected this behaviour, allowing for the avoidance of nonlinearities and offering a more tractable problem. The suggested approach assumes a situation where products bought from farmers and placed in chosen warehouses are wholly transferred to current markets and totally sold. In our approach, the longer the products are kept in storage, the more their quality deteriorates, leading to a decrease in selling price.

We implemented the model in an application case for small-scale farmers in the Department of Atlántico, Colombia, where several weaknesses in logistics processes related to ASCP have arisen (Gobernación del Atlántico 2016; Minagricultura 2019a). The modelling includes multiple scenarios with different transport fares, modelling horizons, supply and demand patterns, and pricing curves to address several possible market scenarios.

The remainder of the paper is organised as follows. Section 2 presents a literature review. Section 3 describes in detail the assumptions, parameters, sets, variables, objective function, and constraints of the problem. Section 4 describes the features of the implementation case. Results and discussions are showcased in Sect. 5; in this section, we also present comparisons with benchmarking models. Finally, Sect. 6 proposes conclusions and future research paths.

2 Literature review

Several works on the supply chain design for perishable products have been published. Using Table 1, we provide a summary of relevant research in the field. The research conducted by Dai et al. (2018) proposed an inventory location model for a supply chain with perishable products. Their model included capacity and emissions constraints using fuzzy logic. Eskandari-Khanghahi et al. (2018) solved a multi-objective problem for a supply chain of platelets. The study considered the perishable character of blood and the uncertainty of the supply of blood products. They differentiated between "young" and "old" blood according to the remaining life of the products. The model was able to make location and inventory decisions. Mogale et al. (2019) proposed an optimisation model for the location of facilities across India in a FSC dealing with grain. They addressed the problem, including sustainability concerns (multi-objective approach) and economies of scale. The authors found that trains outperformed trucks in terms of transport costs, and the location of several warehouses would eventually increase the profits through the chain. Perishability was not accounted for in the modelling. Musavi and Bozorgi-Amiri (2017) proposed a multi-objective mathematical model for distributing food products to minimise the costs and CO₂ emissions and maximise the quality of perishable products. The authors tackled strategic and operational decisions simultaneously, looking for optimal solutions. The investigation conducted by Zahiri et al. (2018) presented a multi-objective problem that considered economies of scale, demand and supply uncertainty for a pharmaceutical supply chain. The products had fixed shelf life. The model considered strategic and tactical decisions and was solved using robust optimisation.

Some works are specifically related to agricultural products. de Keizer et al. (2017) designed a logistic network for perishable products with heterogeneous quality decay. The European floricultural sector inspired their research. In their approach, the quality of flowers changed from one stage of the chain to another, deteriorating or keeping it the same (in the best case). Also, the modelled problem accounted for demand variability, location decisions, production decisions, and price sensitivity to

Table 1 Reviewed papers

Authors	Model type	Products	Periods	Suppliers	Customers / Retailers / Markets	Perishable Products	Purchasing/production	Facility location	Distribution	Inventory	Different type of facilities	Uncert	Food products	Real-world app	Obj
Dai et al. (2018)	MINLP	MUL	SNG	MUL	MUL	X	-	X	-	X	-	F	-	-	C
Eskandari-Khanghahi et al. (2018)	MILP	SNG	MUL	MUL	MUL	X	-	X	X	X	-	F-PP	-	X	C-E-JO
Mogale et al. (2019)	MINLP	SNG	MUL	MUL	MUL	-	-	X	X	X	-	-	X	X	C-E
Musavi and Bozorgi-Amiri (2017)	MILP	MUL	SNG	SNG	MUL	X	X	X	X	X	-	-	X	X	C-Q-E
Zahiri et al. (2018)	MILP	MUL	MUL	MUL	MUL	X	X	X	X	X	-	RO-F	-	-	C-UD
de Keizer et al. (2017)	MILP	MUL	SNG	MUL	MUL	X	X	X	X	X	-	-	X	X	NP
Ghezavati et al. (2017)	MILP	SNG	MUL	MUL	MUL	X	X	X	X	X	-	-	X	X	NP
Keshavarz-Ghorbani and Pasan-dideh, (2021)	MILP	SNG	MUL	MUL	MUL	X	X	X	X	X	-	RO	X	X	C-E

Table 1 (continued)

Authors	Model type	Products	Periods	Suppliers	Customers / Retailers / Markets	Perishable Products	Purchasing/production	Facility location	Distribution	Inventory	Different type of facilities	Uncert	Agro/Food products	Real-world app	Obj
Basso et al. (2023)	MILP	SNG	MUL	SNG	SNG	X	–	–	X	–	–	S	X	X	C
Nasrollahi et al. (2023)	MINLP	MUL	MUL	MUL	MUL	–	–	X	X	–	–	F	X	X	C – UD
Gholian-Jouybari et al. (2024)	MILP	SNG	MUL	MUL	MUL	–	X	X	X	X	–	–	X	X	C – E – JO
Li and Zhang (2024)	MINLP	SNG	MUL	MUL	MUL	X	–	X	–	X	–	S	X	–	NP
Aliabadi et al. (2024)	MILP	SNG	SNG	MUL	MUL	X	X	–	–	X	–	–	–	–	E – NP
Our study	MILP	MUL	MUL	MUL	MUL	X	X	X	X	X	X	–	X	X	NP

Uncert, Uncertainty; *App*, Application; *Obj*, Objectives; *MINLP*, Mixed integer non-linear programming; *MILP*, Mixed integer-linear programming; *MUL*, Multiple; *SNG*, Single; *F*, Fuzzy; *RO*, Robust optimization; *S*, Stochastic; *PP*, Possibilistic programming; *E*, Emissions; *NP*, Net profit; *C*, Cost; *UD*, Unmet demand; *JO*, Job opportunities; *Q*, Quality

products with different qualities. They solved the model in a multi-echelon framework and did not account for hubs or production capacities.

Ghezavati et al. (2017) addressed optimising a supply chain for fresh products. The model, applied to the tomato chain in Iran, considered inventory decisions by type of facility, including cooling and refrigerating facilities, location decisions, and price changes according to the variation in product characteristics (namely, ripeness, maturity, and coolness) and harvesting decisions. Nevertheless, the model did not allow for deciding the type of facilities. In other words, cooling and refrigerating facilities were set previously. It only allowed for deciding whether to locate a warehouse at a specific place. Besides, the model did not consider modelling multiple products within the chain. No capacity was set for warehouses, nor was the quantity harvested. Keshavarz-Ghorbani and Pasandideh (2021) designed an agricultural supply chain for perishable products under uncertainty. They formulated a bi-objective model to minimise total costs and reduce CO₂ emissions. The model considered the shelf life of products, inventory, and location decisions. It also involved different sets of warehouse temperatures that impact the perishability rate. However, the authors did not consider multiple products, nor did they consider the possibility of choosing between refrigerated and non-refrigerated facilities.

Recently, Nasrollahi et al. (2023) proposed a bi-objective model aiming to minimize costs and unmet demand. The authors solved the problem in a multi-stage stochastic framework, including strategic and tactical decisions, namely selecting the best cross-dock, determining the flow of products through the network, the number of trucks to use, and the unmet demand. They formulated their model under the cross-docking policy, impeding the storage of inventory. The model was applied to an ASC of non-perishable products. Basso et al. (2023) formulated a model that addressed the problem of bottling and transporting wine to ports in order to minimize inventory. The model was implemented in a Chilean winery in a large-scale problem, which was solved by means of a heuristic leading to sound solutions in polynomial time.

Gholian-Jouybari et al. (2024) modelled an ASC in the context of the coconut industry under a sustainable framework. The model aimed to manage total costs, environmental effects, and job opportunities while considering reverse flows (closed-loop chain). The authors implemented several metaheuristics to solve the problem. Li and Zhang (2024) proposed a two-stage scenario for a resilient AFSC. The first stage of the model focuses on establishing an equilibrium model that accounts for the characteristics of perishable foods, while the second stage of the model builds upon the outcomes of the first stage by addressing the strategic decisions related to the supply chain network design. The model accounts for location allocation and inventory decisions. Aliabadi et al. (2024) studied the design of a supply chain for time-varying items, considering several managerial features such as promotional efforts, credits, and price-dependent demand. Their bi-objective model aimed to minimize gas emissions and maximize the joint profit of retailers and wholesalers. The authors considered deterioration through a three-parameter Weibull distribution, as well as backorders.

There are several works in lot-sizing problems related to perishable products. For instance, Otrodi et al. (2019) proposed a model to solve the pricing and lot-sizing

problem of perishable products in a two-echelon supply chain simultaneously. Likewise, Li and Teng (2018) conducted research involving pricing and lot-sizing problems, considering a complex framework where demand is dependent on several features such as selling price and product freshness. Acevedo-Ojeda et al. (2020) studied the two-level lot-sizing problem for perishable products, considering three different representations of this property by means of a fixed shelf-life, functionality deterioration, and functionality-volume deterioration. However, these studies concentrate solely on tactical choices, disregarding strategic decisions like the facility location issue, thereby limiting their scope to a single tier or level within the supply chain.

The review shows a significant research gap in the formulation and application of mathematical models to optimally design supply chains for perishable products within the context of agricultural products. In this paper, we proposed a mixed-integer linear model to design ASCP. The model is multi-product, multi-period, with capacity constraints that account for strategic (production levels, location of warehouses, allocation of suppliers, and allocation of customers) and tactical–operational decisions (inventory levels). For the analysis, we have considered that since agricultural products are perishable, they lose quality—and therefore value—as they approach shelf life. Our model’s remarkable particularity relies on the type of warehouse set in each location: the model can decide between a *regular* warehouse and a *refrigerated* one (which is different from the cooling efforts addressed in many other publications). The latter decision affects the perishability of products directly, hence the profit, but also represents a different cost to the supply chain according to the decision.

3 Model formulation

This section briefly introduce perishability modelling and the approach employed to solve the problem. Then, we present the sets, parameters, variables and the proposed model formulation. Sets, parameters, and variables are described in Table 2, 3, and 4., respectively. The objective function and the related constraints are presented in Sect. 3.4.

Table 2 Sets of the model

$I = \{1, 2, \dots, I \}$	Set of potential locations for facilities
$J = \{1, 2, \dots, J \}$	Set of markets (final customers)
$M = \{1, 2, \dots, M \}$	Set of suppliers
$P = \{1, 2, \dots, P \}$	Set of products
$T = \{1, 2, \dots, T \}$	Planning horizon of discrete periods
$K = \{1, 2, \dots, K \}$	Set of qualities of products
$R = \{1, 2\}$	Set of types of facilities (refrigerated, not refrigerated)

Table 3 Parameters

C_{pmi}	Transport cost per tonne of product p from supplier m to facility i
C_{pij}	Transport cost per tonne of product p from the facility located in i to market j
C_{lp}	Inventory cost per tonne of product p
C_r	Location cost for the facility of type r
C_{Prod}^{pm}	Production cost per tonne of product p at supplier m
CAP_{ri}	Inventory capacity (tonnes) of the facility type r at location i
D_{pjt}	Demand for product p (tonnes) from client j in period t
$CapOfer_{mp}$	Production capacity of product p (tonnes) at supplier m
V_{pkr}	Selling cost per tonne of product p with quality k coming from facility type r

Table 4 Decision variables

X_{ir}	1 if a facility of type r is set in location i , 0 otherwise
A_{im}	1 if supplier m provides products to warehouse i , 0 otherwise
Y_{ij}	1 if client j is assigned to warehouse i , 0 otherwise
Q_{mpt}	Quantity of product p provided by supplier m in period t
I_{pitk}^r	Inventory level of product p with quality k stored at a facility type r in i , at the end of period t
W_{pijtk}^r	Amount of product p with quality k transported from facility type r located in i to market j in period t
S_{pmit}^r	Amount of product p (always with quality $k=1$) transported from supplier m to a facility type r located in i in period t

3.1 Perishability modelling

Modelling perishability in supply chain network (SCN) design is a highly complex task with increasing interest worldwide (Aazami and Saidi-Mehrabad 2021). The features related to this type of chain add significant complexity to the conception and the solution. We propose a mathematical formulation to address strategic and tactical operational decisions in reasonable computational times under a deterministic environment. We highlight the link between quality loss and product depreciation, which allowed for the modelling of perishability through the selling price of products.

Amorim et al. (2013) defined a perishable product as one whose physical conditions deteriorate notoriously or lose value under the consumer's perception over time. Modelling perishability is complex because it involves temperature, humidity, pressure, light, and the presence of microorganisms. Peleg et al. (2002) proposed to use a Weibull power law to describe isothermal degradation for food quality, depending on temperature and time, given by $q(T, t) = q_0 e^{-b(T)t^{n(T)}}$, where $q(T, t)$ is the quality q for a temperature T and a time t , q_0 is the initial quality, t is a time interval, $b(T)$ and $n(T)$ are coefficients depending on temperature.

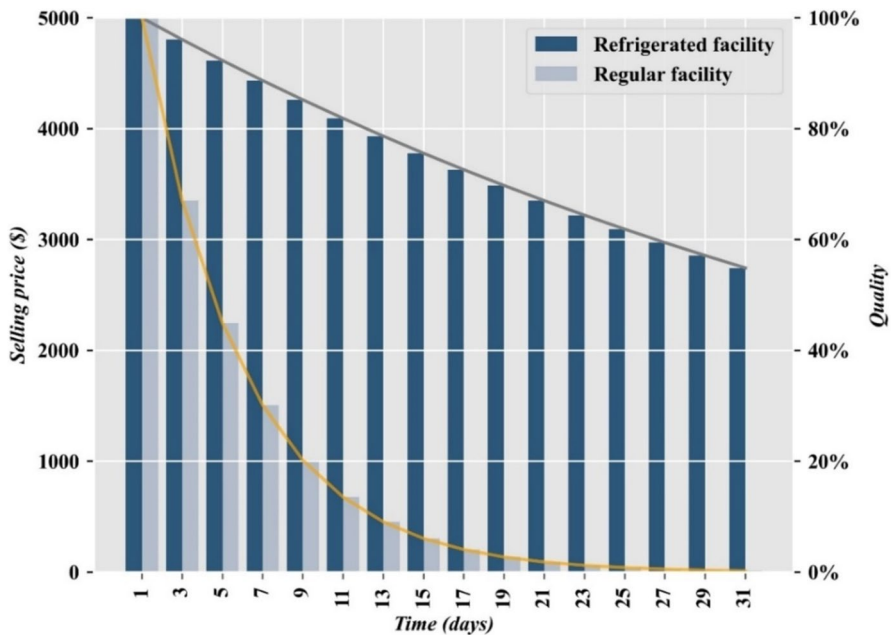


Fig. 1 Discretisation of perishability for fruits

The equation proposed by Peleg et al. (2002) included the effects of time and temperature, which are the main parameters affecting the shelf life of products (Wu et al. 2019). This formulation has been used in several operations research and inventory control studies because it adequately describes the deterioration process (Pahl and Voß, 2014). However, its direct utilisation would imply a non-linear mathematical model, which is computationally more demanding than a linear or mixed-integer linear problem.

A decline in the quality of a perishable product could reduce consumer acceptance (Qin et al. 2014) and lower the product's value over time (Chintapalli 2014). Therefore, we assume that the selling price decreases proportionally with the loss of quality. In addition, the proposed model does not consider time as a continuous variable; instead, time is modelled based on a set of discrete periods (e.g., day by day).

In our approach, product perishability is modelled through product depreciation. We propose a parameter of a range of selling prices, discrete and constant, associated with the deterioration of products. Figure 1. presents the idea graphically: the deterioration curves vary according to the type of facility. At a refrigerated facility, a product preserves its organoleptic and nutritional properties by a longer lapse, keeping its value (higher selling price). However, at regular facilities, the same product in the same lapse degrades rapidly, decreasing its value on the market (lower selling price). Thus, we link the selling price to the quality of the product, which depends on the type of facility it is stored.

Time (orange and grey curves) can be represented by discrete points in time with constant prices (light grey and dark blue bars). Ultimately, this is represented by a

parameter (the selling price parameter) indexed in product type, facility type, and quality (V_{pkr}). Our approach avoids the complexity of including exponential terms in the objective function through a methodological approximation that allows us to treat the problem more efficiently. This way, we shorten a research gap in ASCP design regarding the impact of product freshness on selling prices, as stated by Estes et al. (2021).

Our formulation is similar to the one implemented by de Keizer et al. (2017) in acknowledging the changes in properties of perishable products through time, their impacts on selling prices and, therefore, on revenues. We consider the discretisation of perishability as a practical and tractable approach to model product perishability (Rong et al. 2011). We differ from the previous research in linking the quality discretisation approach with the selling price. We present an example of depreciation for fruits in Fig. 2, both for linear and non-linear types.

3.2 Problem description and assumptions

The supply chain design problem studied assumes that small-scale farmers are coordinated by a supra-organisation that aims to deliver their products, maximising their profit, where involved products are perishable. The organisation intends to design a logistics network to deliver products, which ends with markets. The

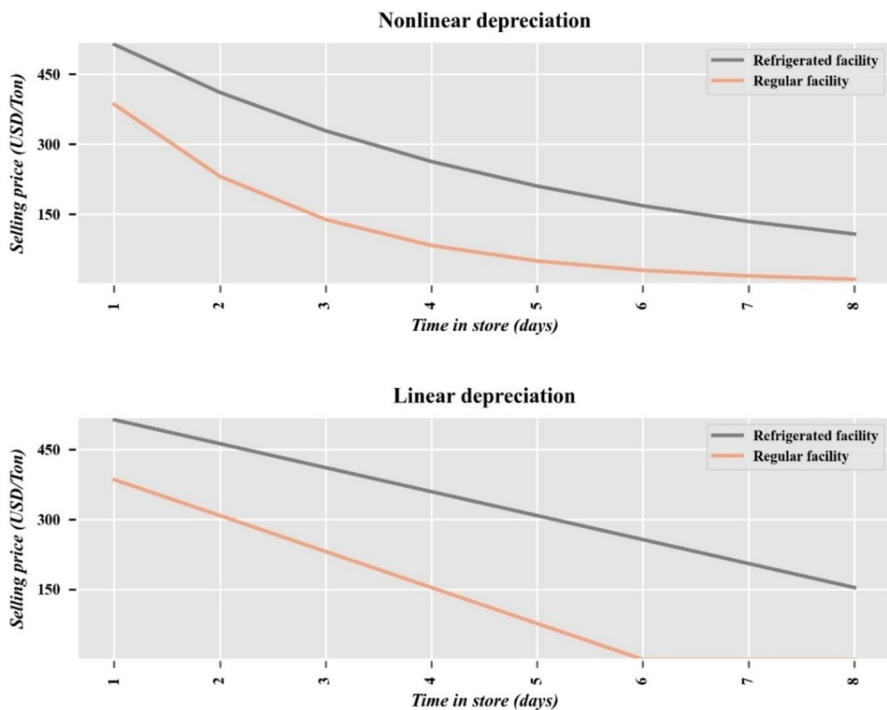


Fig. 2 Depreciation curves for fruits

products are transported from producers (small-scale farmers) to warehouses. Then, they are consolidated and eventually stored to account for temporal offsets between supply and demand. Consolidation has been **stressed** as an appropriate strategy to apply for perishable products (Chen et al. 2018) that may bring benefits regarding sustainability. Moreover, it will allow for taking advantage of economies of scale. The network design considers production costs, fixed costs of facilities (warehouses), transport and inventory costs.

Selling prices to markets are directly related to product quality, which decreases as a function of time and environmental conditions. The supra-organisation aims to assess whether it is convenient to use refrigerated facilities to preserve products better than non-refrigerated ones. However, refrigerated facilities imply higher setting and inventory costs. Figure 3 shows the structure of the modelled supply chain network.

Several assumptions were made in the formulation of the mathematical model:

- All products are delivered to warehouses with maximum quality, which decreases with time.
- The products suffer quality loss over time, which depends on the warehouse (whether refrigerated or not). Consequently, products have different prices according to the type of storage facility for each discrete period of the model.
- The selling price is directly related to the quality, which decreases depending on time and type of facility (refrigerated or regular).
- Facilities are capacitated.
- Supply and demand are fixed and known in advance.

It is worth noting that the proposed formulation assumes a deterministic scenario, where all products purchased from the suppliers (farmers) and placed at selected warehouses are fully transported to the existing markets, and then they are properly and completely sold. Naturally, depending on how long they are

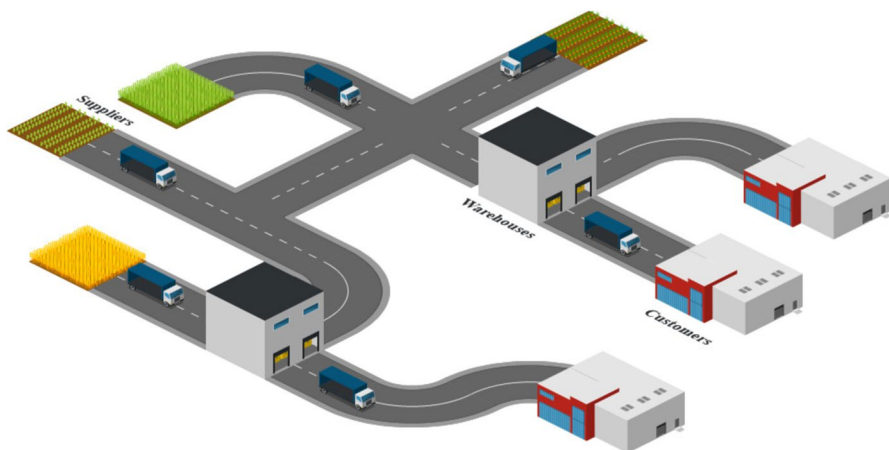


Fig. 3 Structure of the supply chain network

stored in warehouses, products lose quality, and their selling prices are accordingly reduced.

3.3 Sets, parameters, and decision variables

Here, we present the sets, parameters, and variables of the model in Table 2, 3, and 4 respectively.

3.4 Objective function and constraints

$$\begin{aligned}
 \text{Maximize } NP = & \sum_{p \in P} \sum_{i \in I} \sum_{j \in J} \sum_{t \in T} \sum_{k \in K} \sum_{r \in R} W_{pijtk}^r V_{pkr} \\
 & - \sum_{m \in M} \sum_{p \in P} \sum_{t \in T} Q_{mpt} C_{Prod}^{pm} - \sum_{p \in P} \sum_{m \in M} \\
 & \sum_{i \in I} \sum_{t \in T} \sum_{r \in R} S_{pmi}^r C_{pmi} - \sum_{p \in P} \sum_{i \in I} \sum_{j \in J} \sum_{t \in T} \sum_{k \in K} \\
 & \sum_{r \in R} W_{pijtk}^r C_{pij} - \sum_{i \in I} \sum_{r \in R} X_{ir} C_r - \sum_{p \in P} \sum_{i \in I} \sum_{t \in T} \sum_{k \in K} \sum_{r \in R} C_{Ipr} I_{pitk}^r
 \end{aligned} \quad (1)$$

Subject to:

$$\sum_{r \in R} X_{ir} \leq 1 \quad i \in I, \quad (2)$$

$$Y_{ij} \leq \sum_{r \in R} X_{ir} \quad \forall i \in I, j \in J \quad (3)$$

$$A_{im} \leq \sum_{r \in R} X_{ir} \quad \forall i \in I, m \in M \quad (4)$$

$$\sum_{i \in I} \sum_{r \in R} S_{pmi}^r = Q_{mpt} \quad \forall p \in P, m \in M, t \in \{1, \dots, |T|\} \quad (5)$$

$$Q_{mpt} \leq CapOffer_{mp} \quad \forall p \in P, m \in M, t \in \{1, \dots, |T|\} \quad (6)$$

$$S_{pmi}^r \leq CAP_{ri} * A_{im} \quad \forall r \in R, p \in P, i \in I, m \in M, t \in \{1, \dots, |T|\} \quad (7)$$

$$W_{pijtk}^r \leq (D_{pjt}) * X_{ir} \quad \forall p \in P, i \in I, r \in R, j \in J, t \in \{1, \dots, |T|\}, k \in \{1, \dots, |K|\} \quad (8)$$

$$I_{pitk}^r = \sum_{m \in M} S_{pmi}^r - \sum_{j \in J} W_{pijtk}^r \quad \forall p \in P, i \in I, r \in R, t \in \{1, \dots, |T|\}, k = 1 \quad (9)$$

$$I_{pit}^r = I_{pit-1,k-1}^r - \sum_{j \in J} W_{pijk}^r \forall p \in P, i \in I, r \in R, t \in \{2, \dots, |T|\}, k \in \{2, \dots, K\} \quad (10)$$

$$I_{pit}^r = \sum_{k=1}^K I_{pitk}^r \forall p \in P, i \in I, r \in R, t \in \{1, \dots, |T|\} \quad (11)$$

$$\sum_{j \in J} W_{pijk}^r \leq I_{pit-1,k-1}^r \forall p \in P, i \in I, r \in R, t \in \{2, \dots, |T|\}, k \in \{2, \dots, |K|\} \quad (12)$$

$$\sum_{r \in R} W_{pijk}^r \leq D_{pjt} * Y_{ij} \forall p \in P, i \in I, j \in J, t \in \{1, \dots, |T|\}, k \in \{1, \dots, |K|\} \quad (13)$$

$$\sum_{r \in R} \sum_{i \in I} \sum_{k=1, \dots, K} W_{pijk}^r \leq D_{pjt} \forall p \in P, j \in J, t \in \{1, \dots, |T|\} \quad (14)$$

$$\sum_{p \in P} I_{pit}^r \leq CAP_{ri} * X_{ir} \forall r \in R, i \in I, t \in \{1, \dots, |T|\} \quad (15)$$

$$I_{pitk}^r = 0 \forall p \in P, r \in R, i \in I, k \in \{2, \dots, |K|\}, t = 1 \quad (16)$$

$$W_{pijk} = 0 \forall p \in P, i \in I, j = J, t = 1, k \in \{2, \dots, |K|\} \quad (17)$$

$$\sum_{j \in J} W_{pijk} \leq \sum_{m \in M} S_{pmikt} \forall p \in P, i \in I, t \in \{1, \dots, |T|\}, k \in \{1, \dots, |K|\} \quad (18)$$

$$Q_{mpt}, S_{pmikt}^r, I_{pitk}^r, I_{pit}, W_{pijk}^r \geq 0 \forall m \in M, p \in P, i \in I, r \in R, j \in J, t \in \{1, \dots, |T|\}, k \in \{1, \dots, |K|\} \quad (19)$$

$$X_{ir}, A_{im}, Y_{ij} \in \{1, 0\} \forall i \in I, m \in M, j \in J, r \in R \quad (20)$$

Expression (1) is the objective function (O.F.), i.e., maximising the net profit (NP). The first term represents the revenue associated with products sold at the markets. The second is the production costs at the suppliers. The third and fourth terms are transport costs, from suppliers to warehouses and from warehouses to markets. The fifth term stands for the total fixed cost of setting facilities considering both types of facilities (regular and refrigerated).¹ The last term represents the inventory costs.

¹ We assume that the supply chain does not build the facilities and rather uses current facilities, reason why leasing and adequacy costs must be paid.

The set of Eqs. (2) guarantees that, at most, one type of facility can be set at a specific location. Consistency constraints (3) and (4) ensure that suppliers and markets are assigned only to open facilities. Constraint (5) guarantees that, for each specific period, the total amount of a product sent from a supplier to all assigned warehouses is, at most, equal to the production of that product from that specific supplier. Constraint (6) ensures that a supplier cannot deliver more than its production capacity. Constraint (7) states that the amount of a product sent from a supplier to a specific facility is at most equal to the facility's capacity. At the same time, Eq. (8) avoids shipping products from a not open facility.

Equation (9) establishes the inventory balance of a product with maximum quality ($k=1$) at a facility for each period: it should be equal to the total amount of product with maximum quality incoming from all suppliers in that period, minus the total amount of product immediately sent to all assigned markets at the same period. Similarly, constraint (10) states the inventory balance for products with a quality different than the maximum ($k \neq 1$). The inventory level of a product with quality worse than the best at a facility for each period t should equal the remaining inventory at the previous period $t-1$ minus the amount of the product sent to all assigned markets.

Constraint (11) computes the total inventory (I_{pit}^r) for each period, at every facility, product, and facility type ($r=1, 2$). Meanwhile, constraints (12) state that the amount of each product p transported to all markets at every period t ($t \leq 2$) and for each quality k , cannot be greater than the respectively available inventory at the previous period ($t-1$), with a quality $k-1$, for each warehouse and each type of facility ($r=1, 2$).

Equation (13) ensures that sending products from warehouses satisfies assigned markets' demand. At the same time, constraint (14) states that the maximum quantity shipped from all warehouses to each market j , product p and period t , is at most the respective demand, D_{pjt} . Restriction (15) ensures that the total inventory should not exceed the facility's capacity. Constraint (16) warrants that there is no inventory at the beginning of the rolling horizon. Constraint (17) avoids dispatching in other than maximum quality ($k \leq 2$) at the beginning of the rolling horizon. Equation (18) ensures that the amount sold does not exceed the amount supplied for each product, facility, and quality. Constraints (16)–(18) aid the model in preserving consistency and coherence. However, they can be relaxed in other kinds of problems or applications. Finally, constraints (19) and (20) state the domain for decision variables.

4 Case study

In this section, we briefly describe the characteristics of implementing the case study and explain further assumptions made for the case.

4.1 Modelling context

We implemented the proposed model for a case in the department of Atlántico, Colombia. At least 50% of Atlántico's land is suitable for cultivation (IGAC 2016),

and the local and national policies strive towards enhancing agriculture as a pillar of the Colombian economy. The climate is tropical, with an average temperature of 28 °Celsius and relative humidity of 80%. This condition strongly affects agricultural products.

Authorities in the region are concerned about the poor logistical conditions small farmers face. Firstly, the low level of technicality of the production process severely affects small farmers. Their economic burden is stressed because of the access to raw materials and transport. Also, farmers have a low purchasing power, and fertiliser and most other raw materials are expensive—imported products. On the other hand, it has been reported that the department of Atlántico requires logistics infrastructure to support cold chains, where proper handling of agricultural products would reduce losses and preserve their quality. Specifically, the department lacks enough facilities to consolidate and store agrarian food products. Currently, there are only two centres (facilities) for such purposes in the whole department. Neither have sanitation, hygiene, and safety conditions to handle perishable agricultural products nor is there enough capacity (Minagricultura 2019a, p. 41).

We applied our model to perishable agricultural products in the department of Atlántico. The application to a real case helped bring methodological approximated solutions to tackle the problem faced by small-scale farmers and the whole ASCP.

4.2 Modelling scenarios and assumptions for the implementation case

We modelled multiple scenarios. Not only were static values changed, but the form of curves (such as the curve of supply) was also varied. Each scenario had a different combination of the parameters described in Table 5. According to the full factorial experimental design, a total of 64 scenarios were modelled. The purpose was to assess several modelling contexts to identify trends and the impact of specific variations in the network design or some variables (e.g., inventory level).

Colombia, a tropical country, has no seasons. Thanks to the influence of weather conditions, production is highly variable for small-scale farmers. The harvesting season for agricultural products in the Atlántico department typically takes two or three months, which may vary depending on the product. Therefore, we assumed a planning horizon among 60 and 90 periods to represent this behaviour. Supply and

Table 5 Parameters to vary for modelling scenarios

Factor (parameter)	Levels	Description of levels
Supply curve	2	Growing, Ramp
Demand curve	2	Growing, Ramp
Transport fare	2	Low, High
Availability of demand	2	Completely available in horizon, Partially available in horizon
Depreciation curve	2	Linear depreciation, Non-linear depreciation
Horizon	2	60 periods (days), 90 periods

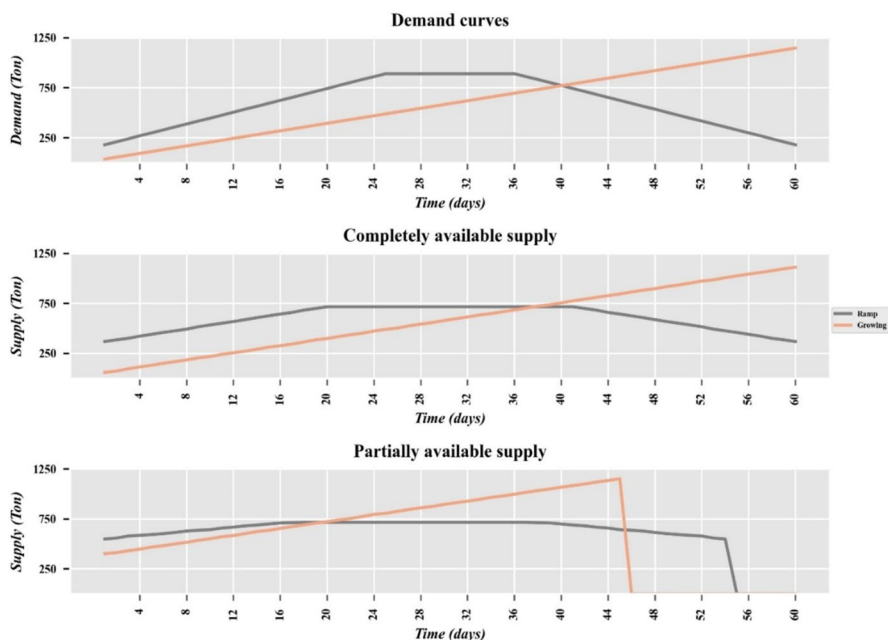


Fig. 4 Supply and demand curves

Table 6 Number of suppliers, markets, and possible locations for facilities

	Number
Suppliers	21
Markets	6
Possible locations for setting up Facilities	21

demand curves are depicted in Fig. 4. Supply data was taken from historical records (at least ten years) of the National Agriculture Ministry (Minagricultura 2019b).

Unfortunately, up-to-date and exact information about the demand for each product in the modelling context was not available. We derived an estimated demand from the national survey of nutritional conditions (Samper et al. 2005) that defines the amount of fruit and vegetable consumption by Colombian citizens.

Table 6 states the number of suppliers, markets, and possible locations for setting facilities. These values were the same for every instance evaluated. The location of each is showcased in Fig. 5.

Moreover, the parameters used for the case study are displayed in Table 7, 8, 9, and 10. The values are the results of market research. The prices, originally in COP, were converted to dollars using the exchange rate of 1 USD = 3700 COP.

Production costs were taken from an official report from Ministerio de Agricultura y Desarrollo Rural and Universidad Sergio Arboleda (2018).

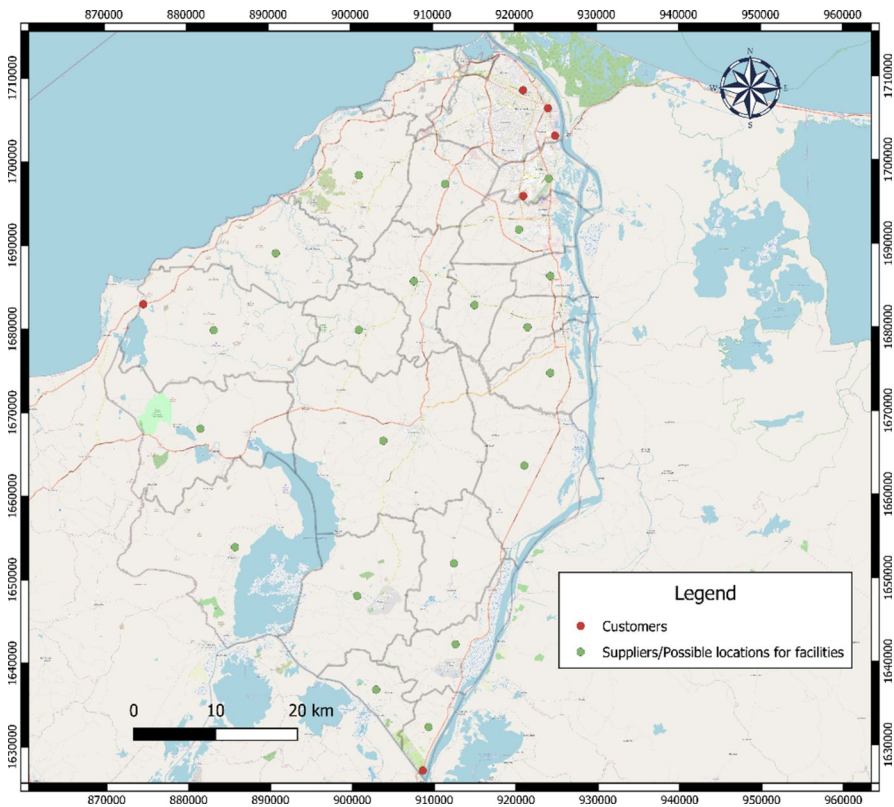


Fig. 5 Location of suppliers, potential warehouses, and markets

Table 7 Transport costs

Origin—Destination	Low (USD/ton-km)	High (USD/ton-km)
Supplier—Warehouse	0.153	0.191
Warehouse—Markets	0.136	0.109

Table 8 Production costs

Product	Production cost (USD/ton)
Fruits	158.9
Vegetables	209.5

Facility setting costs were estimated through official information from DANE (2012) and direct consultations with agricultural supply centres.

Table 9 Facility Setting Costs

Time horizon (days)	Regular facility (USD)	Refrigerated facility (USD)
60	18,456.8	26,543.2
90	27,685.1	39,814.9

Table 10 Selling prices and inventory costs

Product selling price		
Product	Regular facility (USD/ton)	Refrigerated facility (USD/ton)
Fruits	385.7	514.3
Vegetables	287.3	359.1
Inventory costs		
Fruits	77.2	102.9
Vegetables	57.5	71.8

Selling prices were taken from the official report from *Sistema de Información y Precios – SIPS* (DANE). We considered a period of 16 months (December 2018 to April 2020) to estimate the mean selling price of vegetables and fruits.

Several additional assumptions were made for the case study. Those assumptions are as follows:

- There are two types of facilities: regular and refrigerated,
- Regular warehouses offer higher capacity than refrigerated ones,
- There are two types of products: fruits and vegetables,
- Products are in the harvest season,
- Suppliers can provide products every day during the modelling horizon,
- Inventory cost is set to 20% of the nominal selling price, and
- For maximum quality ($k = 1$), fruits in regular warehouses have a selling price for fruits equal to 75% of the same product in a refrigerated warehouse; analogously, 80% for vegetables.

A notable element of our study is the consideration of two depreciation curves (linear and non-linear). Thus, we intended to represent products that perish differently (some quicker than others). For instance, we can mention that some fruits, like mangoes, do not deteriorate as quickly as bananas. Clearly, the non-linear depreciation curve stands for products with an aggressive loss of quality; meanwhile, the linear depreciation curve stands for products with a less aggressive loss of quality.

Table 11 Features of instances chosen

Model instance	Supply	Demand	Horizon	Availability of supply	Depreciation	Transport fares
1	Growing	Growing	60	Completely available	Linear	High
2	Ramp	Ramp	60	Partially available	Linear	High
3	Ramp	Ramp	60	Completely available	Non-linear	High
4	Ramp	Growing	60	Partially available	Non-linear	Low

5 Results

The mixed-integer linear programming model exploited the benefits of simultaneously making strategic and tactical–operational decisions in the complex supply chain network design (Miranda et al. 2009; Orozco-Fontalvo et al. 2019). We were particularly interested in determining the effect of perishability (represented by depreciation) on inventory levels and SCN configuration.

The problem was coded in AMPL® and solved using CPLEX® in a laptop with Intel® Core (TM) i5-420U processor (CPU @ 1.70GHz – 2.4 GHz) and 8 GB of RAM. Despite the thousands of variables and constraints, scenarios were solved optimally in reasonable computational time (up to 2.5 h).

5.1 Model instances

Due to the combination of features showcased in Table 5, 64 instances were evaluated. Our results were similar among the different instances evaluated. For the sake of brevity, we chose four representative instances with features and results approximately equivalent to the others obtained.² Table 11 displays the features of the instances. Table 12 shows the results.

From instance 1, we can see the highest profit of all. This result stems from the form of supply and demand curves. Considering that the supply is entirely available in this instance, we can infer that the model intends to satisfy the demand immediately without storage. We can confirm that by looking at the low inventory costs. Moreover, this instance behaves as a cross-docking model, in which products are transited to warehouses rather than stored. A total of three facilities were set in this instance.

From instance 2, we see the second-highest total profit. Instance 2 differs from instance 1 since supply and demand curves are ramp types here. Further, supply is partially available. Since we modelled deterministic scenarios, the time offset between supply and demand defines whether the model needs to store products. In this case, inventory costs are above 33,000 dollars. The model seeks to satisfy as much demand as possible even if supply is not available at certain moments of the modelled horizon. An interesting feature of instance 2 is regarding the production costs. One could expect the model to use all the production capacity and store the

² Readers may refer to additional data for this paper to check the results of all instances evaluated.

Table 12 Results for instances chosen

Model instance	Objective Function (profit) (USD)	Inventory costs (USD)	Transport costs (suppliers to warehouses) (USD)	Transport costs (warehouses to markets) (USD)	Number of facilities	Location costs (USD)	Production costs (USD)	Solving time (s)
1	19,858,263	671	315,732	165,298	3	77,534	17,057,737	1112
2	17,691,947	33,271	244,381	138,886	3	77,534	15,142,658	871
3	17,543,500	200	239,962	136,647	3	77,534	14,990,105	808
4	14,095,184	1110	154,421	91,625	2	51,689	11,988,974	467

products to satisfy all demands. However, it is more appealing to satisfy a lower percentage of demand with products of higher quality (highly-priced) than to satisfy a higher percentage with lower quality products (which are, as expected, depreciated). We can infer that products are stored for a short time in warehouses, and thus inventory costs are not as high as expected.

Instance 3 comprises a context where the demand and supply curves have the same form (ramp), and supply is entirely available during the modelling horizon. In this case, transport fares are high. The results behave similarly to instance 2. However, in instance 3, the total profit and inventory costs are slightly lower. Given the aggressiveness of non-linear product depreciation, it is understandable that the model is prone to cross-docking.

Instance 4, under a growing demand curve, showing a ramp-type supply (partially available) curve and non-linear depreciation, had the lowest profit. It occurs because of the form of supply and demand curves. Again, the model is prone to a cross-docking logistics system. Because the supply is no longer available, the model stores a minimum number of products to send. In this instance, only two facilities are set. Fewer products to send diminishes the need for storage capacity for the system, which reduces the number of facilities needed.

Figure 6 presents the typical network configuration for the case study. In most cases, three facilities were set, always in the same places: Galapa, Malambo, and

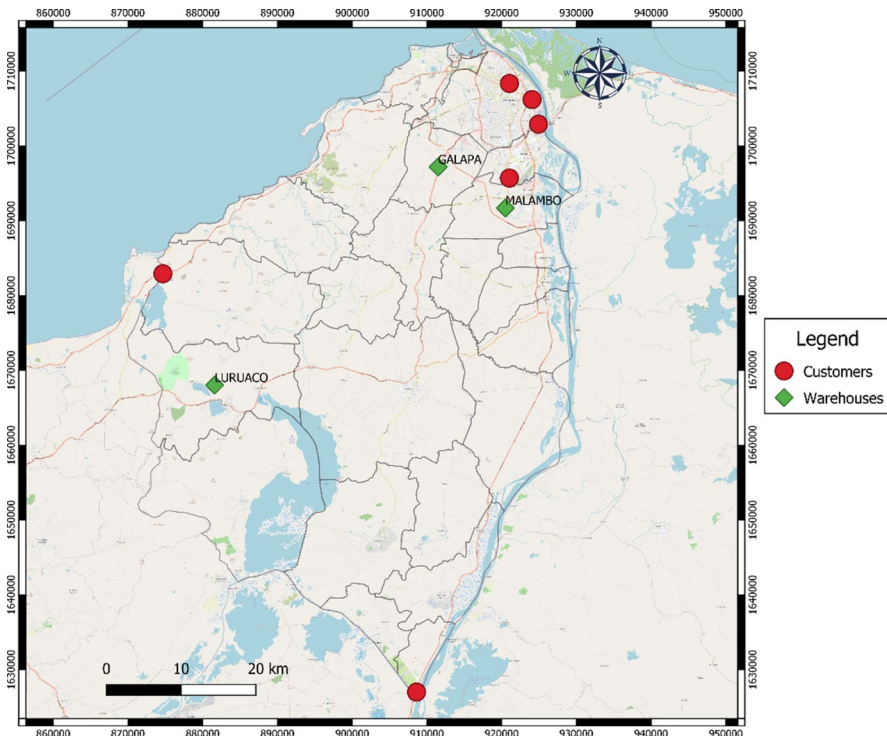


Fig. 6 Typical network configuration

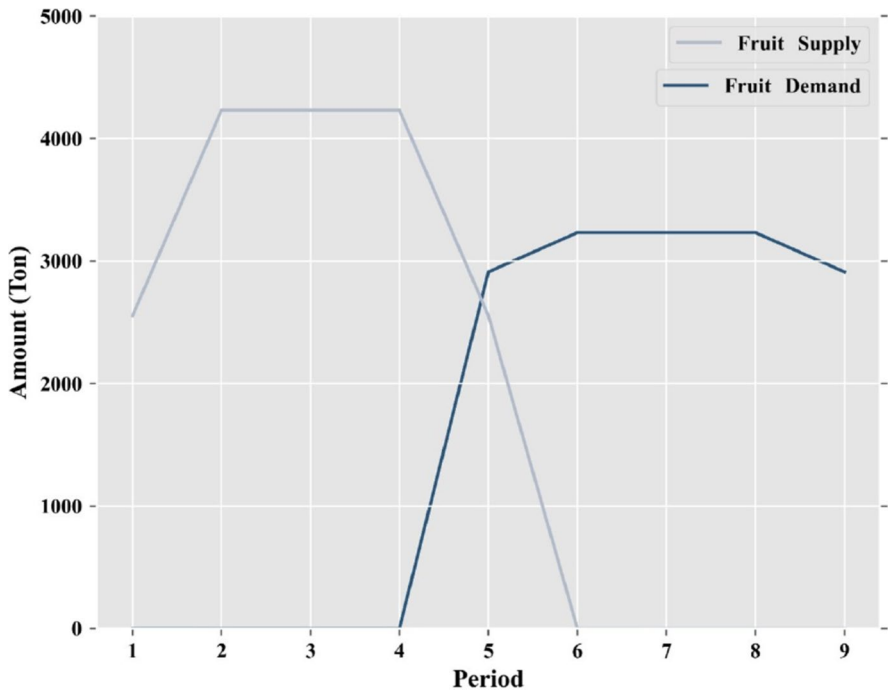


Fig. 7 Supply–demand offset for benchmark model 1

Luruaco. These localities are big suppliers or are close to other big suppliers. The latter explains how the model seeks to reduce the travelled distance in the more costly arcs of the network (from suppliers to warehouses). In those cases where the model chose to set only two facilities, the facility set in Malambo was eliminated, and only Galapa and Luruaco stood. Remarkably, the model set up refrigerated facilities in all cases.

5.2 Benchmark models

The scope of this paper included the evaluation of product perishability and the type of facility to set in an ASCP design. The proposed model was compared against two benchmark models to assess its performance. The modelling context in the benchmark models considered supply and demand with ramp type curves, supply partially available in the modelling horizon, and low transport fares. Mainly, we were interested in testing the impact of perishability in the solutions and the impact of the type of facility (refrigerated or regular) to set. Benchmark model 1 assumes that the products do not perish; consequently, the selling prices are constant. Benchmark model 2 considered products' perishability but was constrained to the use of regular warehouses only.

Table 13 Comparison against benchmark model 1a

Results (USD)	Benchmark model 1a	Correction to profit for benchmark model 1a	Proposed model 1a
Total profit (O.F.)	2,767,342	−650,732 (−76%) ^a	2,419,371
Revenue	6,227,263 (+51%) ^a	4,110,662	5,093,921
Total demand for fruits	25,212		25,212
Total demand for vegetables	45,360		45,360
Total production of fruits	7200		4728
Total production of vegetables	7497		7937
% of fruit demand satisfied	28.6%		18.8%
% of vegetable demand satisfied	16.5%		17.5%
Transport cost (supplier to warehouse)	19,952		37,128
Transport cost (warehouse to markets)	36,457		19,546
Location – setting cost	377,392		206,758
Number of facilities	21		8
Type of facilities	Regular		Refrigerated
Inventory cost	382,945		60,580
Production cost	2,643,184		2,350,537

^aObjective function calculated on the basis of the corrected revenue

^aIn parenthesis, overestimation in revenue compared to real selling price because of naïve assumption regarding depreciation

5.2.1 Benchmark models without depreciation

We intended to estimate the impact of perishability in the ASCP. This section presents benchmark models based on the naïve assumption that products do not perish. According to that assumption, the product selling price is constant in these benchmark models. Other features (e.g., transport, production and inventory costs) and assumptions were held.

To evaluate the performance of the proposed model against the benchmark model, we created a scenario with a shorter time horizon ($t=9$) in which the time offset between supply and demand is stressed. The latter feature can be observed in Fig. 7.

Table 13 showcases the results of benchmark model 1a (a model in which capacities for facilities are the same as the other models) and a model that effectively considers depreciation. On the other hand, Table 14 presents the results for a second benchmark model (1b), keeping the same assumption and structure mentioned above but doubling the capacity for each type of facility.³

³ We hypothesized that, since selling price would be constant, the inventory levels would arise. Therefore, in the first benchmark models we increased the capacity of the facilities.

Table 14 Comparison against benchmark model 1b

Results (USD)	Benchmark model 1b	Correction for profit for benchmark model 1b	Proposed model 1b
Total profit (O.F.) (-107%)	2,896,316 2,419,208	-207,851	
Revenue	7,541,842 (+ 70%) ^a	4,437,659	5,093,921
Total demand for fruits	25,212		25,212
Total demand for vegetables	45,360		45,360
Total production of fruits	9825		4728
Total production of vegetables	7497		7937
% of fruit demand satisfied	39%		18.8%
% of vegetable demand satisfied	16.5%		17.5%
Transport cost (supplier to warehouse)	17,248		41,050
Transport cost (warehouse to markets)	46,854		15,785
Location – setting cost	754,784		206,758
Number of facilities	21		4
Type of facilities	Regular		Refrigerated
Inventory cost	777,387		60,580
Production cost	3,049,237		2,350,537

Several differences arise. First, the network configuration differs for both benchmark models: 21 facilities are opened, and all are regular. Moreover, the benchmark model uses all the facilities the system has available. Because of the increased number of facilities, the distances from the suppliers (small farmers) to the warehouses decreased. Thus, transport costs in the first link of the network are substantially reduced. The opposite occurs for the transport costs from warehouses to markets.

Compared with the proposed model (which effectively considers depreciation), the benchmark model has a different configuration and uses different facilities. Regular facilities imply a lower setting cost than refrigerated ones. In the benchmark model, refrigerated warehouses do not offer any benefit over regular warehouses since there is no change in the selling price over time. Despite the lower setting cost of regular warehouses, the benchmark model exhibits higher location-setting costs because of the difference in the number of facilities.

The selected warehouse type in the benchmark model also provides a higher capacity. This is reflected in the increase in inventory costs. For benchmark model 1a, the increase is over six times the inventory costs for the proposed model. Regarding benchmark model 1b (Table 14), inventory costs are almost thirteen times more than the proposed model. This occurs because the benchmark models seek to increase the total capacity of the chain, allowing it to hold as many products as possible to satisfy the higher percentage of demand.

Notably, the benchmark and proposed models satisfy approximately the same vegetable demand. The breaking point lies in the percentage of fruit demand that is

satisfied. This stems from the fact that fruits are significantly more valuable (higher selling price) than vegetables.

The benchmark models assume no perishability. The selling price is not affected and remains constant. Therefore, the model can store products in inventory without any side effects over the perceived profit. Nevertheless, the central assumption of the benchmark models is naïve. A correction should be applied to the selling price of those products sold with quality 2 or more. This correction is displayed in Table 13 and 14 in the column *correction to profit*. As can be seen, the benchmark models completely miscalculate the profit. Balance collapses to overestimated profits (51% of the first and up to 70% in the second benchmark model compared to the correction). Remarkably, we note that the *true profit* (i.e., the profit calculated based on the corrected revenue) decreases vertiginously: for benchmark model 1a, the reduction is 76%; meanwhile, for benchmark model 1b, the reduction is 107%. Moreover, in the latter case, the model leads to financial losses.

On the other hand, the proposed models consider depreciation and, with a different network configuration, intend to satisfy as much demand as the modelling conditions allow (i.e., offset between supply and demand). These models tend to cross-dock (as we can see from low inventory costs), maximising revenue and fulfilling demand with products of the maximum quality. The total profit perceived by the proposed models is notably higher than the profit provided by the benchmark models.

The benchmark model 1b offers a higher profit than scenarios considering depreciation. Nonetheless, the model overestimates the selling price for products sold from inventory. It is remarkable to say that this overestimation is related to the storage of a greater number of products (inventory) insofar as the number of products of lower quality sold increases. We confirm the suitability of considering perishability when designing ASCP networks.

Table 15 Comparison against benchmark model 2

Results (USD)	Benchmark Model 2	Model with linear depreciation	Model with non-linear depreciation
Total profit (O.F.)	10,340,079	17,767,368	17,753,605
Revenue	25,829,729	33,328,667	33,306,520
Transport cost (supplier to warehouse)	193,019	196,055	195,616
Transport cost (warehouse to markets)	111,129	111,781	111,996
Location – setting cost	53,913	77,534	77,534
Number of facilities	3	3	3
Type of facilities	Regular	Regular	Regular
Inventory cost	19,431	33,271	28,664
Production cost	15,112,158	15,142,658	15,139,105

5.2.2 Benchmark model constrained to regular facilities only

We also compared our model with a benchmark model with linear depreciation but constrained to using regular facilities. In these benchmark models, perishability is considered. This model assumes a time horizon of 60 periods, demand and supply curves with ramp form, and supply that is partially available for the modelling horizon. Table 15 displays the results. The comparison showcases similar values regarding transport costs, the number of facilities and production costs. However, the total profit of the benchmark model is clearly lower than the models with linear and non-linear depreciation. Analysing the transport and production costs, we can see that increasing these costs does not change the total profit. We can infer that there is no evident change in the flows in the network or the quantity produced.

The constraint of using regular facilities explains the changes in the total profit. Regular warehouses are not equipped to preserve product quality as refrigerated warehouses are. Exposure to inappropriate environmental conditions decreases product quality and, accordingly, selling price.

Changes in inventory costs are evident as well. A detailed look at inventory levels can be seen in Fig. 8. Benchmark model 2 keeps low inventory levels, even for the last period with available supply (period 54). It is worth recalling that regular warehouses offer a higher storage capacity than refrigerated warehouses.

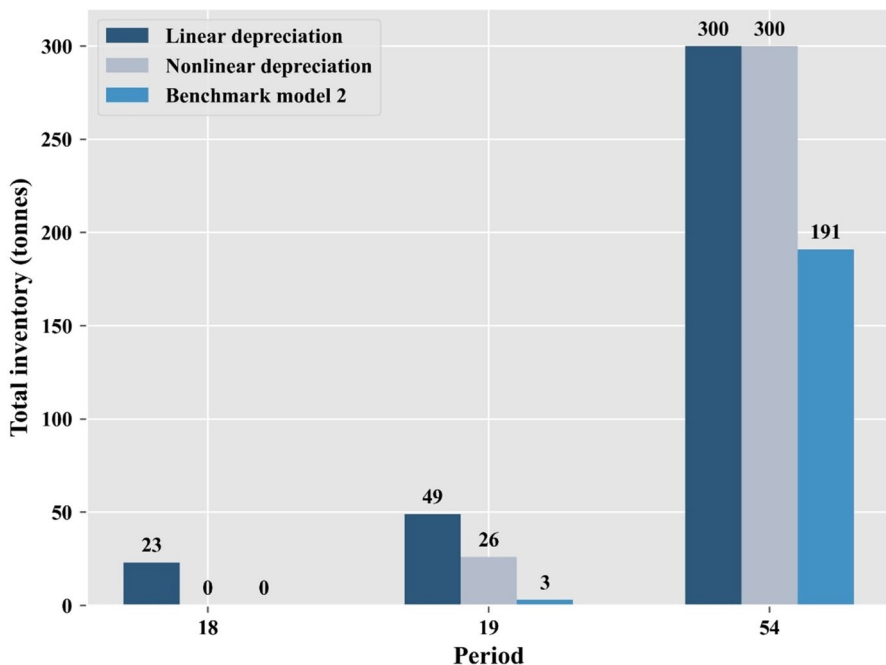


Fig. 8 Comparison of inventory levels for benchmark model 2 and instances with linear and non-linear depreciation

Surprisingly, the model constrained to regular facilities prefers not to satisfy higher demand. We can infer that even when the network can store more products, this is not attractive because of the loss of quality, reflected directly in the selling price and, ultimately, in the perceived profit.

5.3 Discussion

It is worth noticing that the studied problem exhibits a strategic-tactical nature, usually requiring an annual resolution and potentially laying the groundwork for future periods. However, modelling product deterioration necessitates a daily time unit, as inventory arrival times at warehouses lead us to create distinct product states (i.e., quality levels). In addition, the problem is focused on a single season, reflecting the agricultural context, where supply chain planning encompasses warehouse location, purchasing, and transport decisions for the entire season (e.g., 2–6 months).

The proposed model includes a price parameter that embraces the quality of a product as a function of the type of facility and storage time. The formulation is quite flexible when modelling products with different behaviours. For instance, we could model strawberries, which are sold with maximum quality, and after that point, their valuation descends quickly. We could also model products whose valuation follows a triangular function, like mangoes, which can be sold way before complete maturation with an initial price or when the product matures at a higher price that then descends.

However, determining the quality of food products might be a subtle task, as it is susceptible to several factors that might be correlated. Further, we should consider that the changes in properties of perishable products are also subject to the genetics of such products. For small-scale farmers, where the level of technicality is low, the product may behave with moderate variability, even in the same environment or storing conditions. Therefore, our formulation still succeeds in representing this highly complex phenomenon.

From a market perspective, quality is a fundamental topic for the ASCP. Each day, it is more common to see specific requirements for products. This has turned into acceptance or rejection conditions. The latter increases the exigence of the supply chain, narrows the margin of responsiveness in case of contingencies (e.g., disruptions), and stresses the requirements for suppliers and management of products in the middle layers of the chain (warehouses). In our formulation, we did not account for specific quality in demand. We linked the *dissatisfaction* of users when receiving products with lower quality by reducing the selling price of products (at markets). Future research may imply stricter modelling constraints related to this, which reflects more realistic conditions.

Bearing in mind that the results of the benchmark models depend on the modelling conditions and the features considered, we see that not considering the perishability of products in ASCP network design leads to mistaken solutions as it directly impacts profits (Li et al. 2022). Strategic decisions severely impact supply chain costs and functioning. The benchmark models collapse to a configuration where all the system capacity is used. This could eventually lead to even greater losses if the

expected profit is not satisfied (as in benchmark model 1a) or if the chain is not at all financially sustainable (as in benchmark model 1b). In these cases, there is a possibility of larger losses stemming from unused or removed facilities if the chain manager or investors (e.g., government) decide to decline and not support this market anymore, reinforcing the well-known effect of strategic decisions in the long term (Rekabi et al. 2021).

From a methodological perspective, we acknowledge that there is a clear relationship between the proposed formulation in this paper and the widely studied lot-sizing problem (Manne 1958; Wagner and Whitin 1958; Lu et al. 2012; Melega et al. 2020). In both cases, the objective is to determine the quantities to be produced or requested from a supplier during each period, along with related inventory decisions, considering deterministic demands to be fulfilled (Basnet and Leung 2005). Particularly, our model resembles the extension of the lot-sizing problem that addresses strategic decisions such as facility location, similar to (Romeijn et al. 2010; Daluie et al. 2014; Alegoz and Yapicioglu 2022). However, our model and the latter papers differ from pure lot-sizing models in the fact that the amount of product to be fulfilled for each warehouse is not fixed, now depending on customers or markets assigned to each warehouse. In addition, total amounts requested at the warehouses on each period have to be sourced from a set of suppliers. Finally, the most significant novelty of this research relies on integrating product deteriorating into a joint warehouse location with sourcing selection, market assignment, inventory decisions, and warehouse type selection (i.e., refrigerated or regular warehouses).

5.4 Managerial insights

Some managerial insights arise beyond the modelling results and the comparisons with the benchmark models. First, results from Table 13. portray that adequately accounting for perishability significantly impacts the supply chain network design (Esteso et al. 2021; Hosseini-Motlagh et al. 2021). Naïve assumptions over the perishability of products may mislead the network configuration. Particularly, overestimations on the number of facilities to set are translated into high facility-setting costs and, eventually, unused capacity. Once made, this is a severe mistake because facility location decisions are tough to change.

Second, perishability remarkably affects total profit. If loss of quality of perishable agriculture products is not considered, severe overestimations in the total profit may occur. The latter is in line with previous research (Chen et al. 2022). The impact of a misguided management policy is accentuated depending on the type of perishability of a product. Moreover, it is worth highlighting that we worked under the assumption of deterministic demands. In this deterministic scenario with fixed and known market demands, purchasing products that will eventually perish at the located warehouses has no sense since it only increases the total purchasing and transport costs without yielding any benefit from selling. Thus, the proposed formulation does not explicitly consider product perishability but instead involves time-dependent quality worsening and

selling price reduction, considering discrete periods. Modelling product perishability may be reasonable if market demands or quality deteriorating, among other features, entail uncertainty, where there exists a non-zero probability of products staying at a warehouse long enough to reach a null selling price, or buyers in the market are not willing to purchase poor quality products. These issues are beyond the scope of this research and are interesting and promising considerations for future work.

The deterministic condition is known to be hardly fulfilled, and ASCPs are subject to variability. Hence, the more variability in demand, the greater the amount of safety stock (Miranda and Garrido 2004), which may substantially impact the supply chain network configuration and the estimated profit (because of the holding costs related to inventory). Moreover, the supply chain may incur wasting costs because of inedible, disposed products if perishability is not treated or considered correctly from the beginning. Disruptions caused by large-scale may incorporate uncertainty and stress the losses.

Third, the synchrony between supply and demand strongly influences the network structure. Perfect timing allows the product to be shipped directly and immediately from producers to consumers. In this case, a cross-docking logistics model will be the most convenient, almost neglecting the need for storage or refrigeration. However, these cases are exceptional. Therefore, it is convenient to evaluate different scenarios considering various demand and supply patterns over time.

We found that the type of facility to set in a location impacts the quality of handled products noticeably. Consequently, total profit is also affected. In ASCP design, supply chain managers should consider setting facilities with appropriate conditions to store and handle products (Esteso et al. 2021). This could better preserve the products' properties and offer a higher quality to the final consumer (Esmizadeh et al. 2021). The latter is reflected in higher revenues from the products sold. Another insight is that not considering the effect of the type of facilities may lead to setting facilities unsuitable for storing and handling perishable products. Supposed changes in the inventory policy (for example, switching from a traditional inventory system to a cross-docking system) are made after wrong strategic decisions. In that case, the chain could yield a low inventory/total capacity rate, which might not be desirable.

Finally, the model can be adjusted to consider the variability in the quality of supplied products and their impacts on inventory management. The strategy proposed by Coelho and Laporte (2014) or Giallombardo et al. (2022) might be beneficial to planning inventory policies properly.

6 Concluding remarks

This paper presented a mixed integer linear programming model for the design of ASCP. The model considers production, facility location, suppliers' and markets' allocation, and inventory decisions in a multi-period, multi-product framework. The relation between strategic and tactical decisions (as we modelled in this paper) is

complex and demanding in terms of computational resources for solving the proposed model. However, solving them simultaneously is critical for the decision-making process in SCN design. The model includes additional features compared to traditional facility location problems, such as production constraints for suppliers and inventory capacity constraints for warehouses, two types of perishability (linear and non-linear) and two types of facilities to be set (regular and refrigerated). The features of the proposed model constitute one of the main contributions of this research from a methodological perspective. A high number and variety of instances were modelled.

The model was applied in an implementation case for agricultural food products (fruits and vegetables) in the department of Atlántico, Colombia. In all cases, the model chooses refrigerated facilities because of the evident improvement in product storage conditions. Also, the model either tends to use cross-docking or is prone to keeping low inventory levels for a short time. This way, the proposed system aims to deliver products with the highest quality possible, maximising the total profit.

The remarkable contributions of our paper include the type of facility to be set with a direct implication of this decision not only in supply chain network design configuration but also in product perishability and related inventory management. Likewise, we proposed a practical way (through a parameter, avoiding nonlinearities) to model perishability and associate it with the selling price of products. Further, despite the application to a context of perishable agricultural products, the model is sufficiently general to be applied to other modelling contexts that deal with perishable assets, such as the pharmaceutical industry or blood products.

Beyond the modelling results of this research, we draw some managerial implications. It is noteworthy to mention the powerful impact of perishability over strategic decisions and profit estimations. Similarly, there is a potential impact on the rate of inventory and unused capacity of facilities if perishability is not accounted for properly. Further, we remark on the incidence of the type of facility to set in an ASCP. Suitable facilities (those that provide appropriate environmental conditions) may considerably slow down the degradation process inherent to perishable products. Meanwhile, unsuitable facilities may severely impact total profit and potentially cause the chain to incur wastage costs.

This paper made several assumptions to make the problem more tractable. First, we assumed that both demand and supply are known and deterministic. To account for this critical topic, we suggest that future research may account for variability or uncertainty in demand and supply. Stochastic programming and robust optimisation are two modelling approaches suitable for such a purpose. Second, we assumed that all products are supplied with maximum quality, i.e., no supplied product with any imperfection. Agricultural food products can be sensitive not only to environmental conditions but also to handling and storing conditions. Hence, we believe another interesting research path should consider the variability in the quality of supplied products. This inherently represents a challenge regarding inventory policies. Third, we did not consider the possibility of capacity transfer (flow of products) between warehouses. Not accounting for it could lead to wastage when dealing with perishable products. Thus, another research area may consider a capacity transfer in ASCP design.

We aimed to model an ASCP for small-scale farmers with scarce levels of technicality and purchasing power. Therefore, a fourth extension may strive to include equity in modelling by guaranteeing that all suppliers (farmers) have a facility within a determined range (*radius*) to minimise transport costs. We should also remember that we proposed a mixed integer linear formulation. However, perishability is often better described by non-linear expressions. Solving a non-linear model to optimality can be a challenge for further research. Finally, the integration between different seasons in a joint model, where various conditions and decisions may dynamically evolve across the seasons, thus sharing the initial investment decision (warehouse location), represents significant future research.

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Data availability The data that support the findings of this study are available from the corresponding author upon reasonable request.

Declarations

Conflict of interest The authors have no conflict of interests related to this publication.

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References

- Aazami A, Saidi-Mehrabad M (2021) A production and distribution planning of perishable products with a fixed lifetime under vertical competition in the seller-buyer systems: a real-world application. *J Manuf Syst* 58:223–247. <https://doi.org/10.1016/j.jmsy.2020.12.001>
- Acevedo-Ojeda A, Contreras I, Chen M (2020) Two-level lot-sizing with raw-material perishability and deterioration. *J Oper Res Soc* 71:417–432. <https://doi.org/10.1080/01605682.2018.1558942>
- Afshar MA, Hosseini SMH, Sahraeian R (2022) A bi-objective cold supply chain for perishable products considering quality aspects : a case study in iran dairy sector. *Int J Eng* 35:458–470. <https://doi.org/10.5829/jje.2022.35.02b.22>
- Ahumada O, Villalobos JR (2009) Application of planning models in the agri-food supply chain: A review. *Eur J Oper Res* 196:1–20. <https://doi.org/10.1016/j.ejor.2008.02.014>
- Alegoz M, Yapicioglu H (2022) Integrating supplier selection, lot sizing and facility location decisions under a TBL approach: a case study. *Soft Comput* 26:4819–4830. <https://doi.org/10.1007/s00500-022-06866-7>
- Aliabadi L, Zegordi SH, Husseinzadeh Kashan A, Rastegar MA (2024) A sustainable supply chain model for time-varying deteriorating items under the promotional cost-sharing policy and three-level trade credit financing. *Oper Res* 24:17. <https://doi.org/10.1007/s12351-024-00824-x>

- Amorim P, Günther HO, Almada-Lobo B (2012) Multi-objective integrated production and distribution planning of perishable products. *Int J Prod Econ* 138:89–101. <https://doi.org/10.1016/j.ijpe.2012.03.005>
- Amorim P, Meyr H, Almeder C, Almada-Lobo B (2013) Managing perishability in production-distribution planning: a discussion and review. *Flex Serv Manuf J* 25:389–413. <https://doi.org/10.1007/s10696-011-9122-3>
- Baloch MK, Bibi F (2012) Effect of harvesting and storage conditions on the post harvest quality and shelf life of mango (*Mangifera indica* L.) fruit. *South African J Bot* 83:109–116. <https://doi.org/10.1016/j.sajb.2012.08.001>
- Baloch MK, Bibi F, Jilani MS (2011) Quality and shelf life of mango (*Mangifera indica* L.) fruit: As affected by cooling at harvest time. *Sci Hortic (Amsterdam)* 130:642–646. <https://doi.org/10.1016/j.scienta.2011.08.022>
- Basnet C, Leung JMY (2005) Inventory lot-sizing with supplier selection. *Comput Oper Res* 32:1–14. [https://doi.org/10.1016/S0305-0548\(03\)00199-0](https://doi.org/10.1016/S0305-0548(03)00199-0)
- Basso F, Contreras JP, Pezosa R et al (2023) Optimizing the wine transportation process from bottling plants to ports. *Oper Res* 23:37. <https://doi.org/10.1007/s12351-023-00778-6>
- Calleja G, Corominas A, Martínez-Costa C, de la Torre R (2018) Methodological approaches to supply chain design. *Int J Prod Res* 56:4467–4489. <https://doi.org/10.1080/00207543.2017.1412526>
- Chakraborty D, Jana DK, Roy TK (2018) Two-warehouse partial backlogging inventory model with ramp type demand rate, three-parameter Weibull distribution deterioration under inflation and permissible delay in payments. *Comput Ind Eng* 123:157–179. <https://doi.org/10.1016/j.cie.2018.06.022>
- Chen J, Dong M, Xu L (2018) A perishable product shipment consolidation model considering freshness-keeping effort. *Transp Res Part E Logist Transp Rev* 115:56–86. <https://doi.org/10.1016/j.tre.2018.04.009>
- Chen G, Wahab MIM, Fang L (2022) Optimal replenishment strategy for a single-manufacturer multi-retailer cold chain considering multi-stage quality degradation. *Appl Math Model* 104:96–113. <https://doi.org/10.1016/j.apm.2021.11.019>
- Chintapalli P (2014) Simultaneous pricing and inventory management of deteriorating perishable products. *Ann Oper Res* 229:287–301. <https://doi.org/10.1007/s10479-014-1753-9>
- Chopra S, Meindl P (2016) Supply chain management: strategy, planning, and operation, 6th edn. London, Pearson
- Coelho LC, Laporte G (2014) Optimal joint replenishment, delivery and inventory management policies for perishable products. *Comput Oper Res* 47:42–52. <https://doi.org/10.1016/j.cor.2014.01.013>
- Dai Z, Aqlan F, Zheng X, Gao K (2018) A location-inventory supply chain network model using two heuristic algorithms for perishable products with fuzzy constraints. *Comput Ind Eng* 119:338–352. <https://doi.org/10.1016/j.cie.2018.04.007>
- Daluie AS, Kuzgunkaya O, Vidyarthi N (2014) The Impact of Emissions on Lot Sizing Decisions in Multi-Facility Network under Congestion. *IIE Annu. Conf. Proc.* 2771–2779
- DANE Mayoristas Boletín Semanal. <https://www.dane.gov.co/index.php/estadisticas-por-tema/agropecuario/sistema-de-informacion-de-precios-sipsa/mayoristas-boletin-semanal-1>. Accessed 4 Sep 2020
- DANE (2012) Boletín trimestral ABASTECIMIENTO DE ALIMENTOS. In: Caracter. las Instal. y el abastecimiento en Corabastos. <https://www.google.com/search?client=firefox-b-d&q=capacidad+bodegas+dane>. Accessed 27 Nov 2020
- de Keizer M, Akkerman R, Grunow M et al (2017) Logistics network design for perishable products with heterogeneous quality decay. *Eur J Oper Res* 262:535–549. <https://doi.org/10.1016/j.ejor.2017.03.049>
- Eskandari-Khanghahi M, Tavakkoli-Moghaddam R, Taleizadeh AA, Amin SH (2018) Designing and optimizing a sustainable supply chain network for a blood platelet bank under uncertainty. *Eng Appl Artif Intell* 71:236–250. <https://doi.org/10.1016/j.engappai.2018.03.004>
- Esmizadeh Y, Bashiri M, Jahani H, Almada-Lobo B (2021) Cold chain management in hierarchical operational hub networks. *Transp Res Part E Logist Transp Rev* 147:102202. <https://doi.org/10.1016/j.tre.2020.102202>
- Esteso A, Alemany MME, Ortiz Á (2021) Impact of product perishability on agri-food supply chains design. *Appl Math Model* 96:20–38. <https://doi.org/10.1016/j.apm.2021.02.027>
- Galal NM, El-Kilany KS (2016) Sustainable agri-food supply chain with uncertain demand and lead time. *Int J Simul Model* 15:485–496. [https://doi.org/10.2507/USIMM15\(3\)8.350](https://doi.org/10.2507/USIMM15(3)8.350)

- Garcia DJ, You F (2015) Supply chain design and optimization: challenges and opportunities. *Comput Chem Eng* 81:153–170. <https://doi.org/10.1016/j.compchemeng.2015.03.015>
- Ghezavati VR, Hooshyar S, Tavakkoli-Moghaddam R (2017) A Benders' decomposition algorithm for optimizing distribution of perishable products considering postharvest biological behavior in agri-food supply chain: a case study of tomato. *Cent Eur J Oper Res* 25:29–54. <https://doi.org/10.1007/s10100-015-0418-3>
- Gholian-Jouybari F, Hajiaghahi-Keshteli M, Smith NR et al (2024) An in-depth metaheuristic approach to design a sustainable closed-loop agri-food supply chain network. *Appl Soft Comput* 150:111017. <https://doi.org/10.1016/j.asoc.2023.111017>
- Giallombardo G, Mirabelli G, Solina V (2022) Optimal storage and shipment management of perishable agri-products with a hybrid priority policy : a case study. *Int J Manag Sci Eng Manag* 00:1–12. <https://doi.org/10.1080/17509653.2022.2092910>
- Giri BC, Jalan AK, Chaudhuri KS (2003) Economic order quantity model with Weibull deterioration distribution, shortage and ramp-type demand. *Int J Syst Sci* 34:237–243. <https://doi.org/10.1080/0020772131000158500>
- Glock CH, Grosse EH, Ries JM (2014) The lot sizing problem: a tertiary study. *Int J Prod Econ* 155:39–51. <https://doi.org/10.1016/j.ijpe.2013.12.009>
- Gobernación del Atlántico (2016) Plan de desarrollo 2016–2019
- Gustavsson J, Cederberg C, Sonesson U et al (2011) Global food losses and food waste: extent, causes and prevention. *Int Congr save Food* 365:3065–3081. <https://doi.org/10.1098/rstb.2010.0126>
- Hosseini-Motlagh SM, Samani MRG, Abbasi Saadi F (2021) Strategic optimization of wheat supply chain network under uncertainty: a real case study. *Oper Res* 21:1487–1527. <https://doi.org/10.1007/s12351-019-00515-y>
- IGAC (2016) La mitad del Atlántico tiene tierras arables aptas para el desarrollo agropecuario
- Kayikci Y, Subramanian N, Dora M, Bhatia MS (2022) Food supply chain in the era of Industry 4.0: Blockchain technology implementation opportunities and impediments from the perspective of people, process, performance, and technology. *Prod Plan & Control* 33(2–3):301–321. <https://doi.org/10.1080/09537287.2020.1810757>
- Keshavarz-Ghorbani F, Pasandideh SHR (2021) A Lagrangian relaxation algorithm for optimizing a bi-objective agro-supply chain model considering CO₂ emissions. *Ann Oper Res*. <https://doi.org/10.1007/s10479-021-03936-1>
- Lemaire A, Limbourg S (2019) How can food loss and waste management achieve sustainable development goals? *J Clean Prod* 234:1221–1234. <https://doi.org/10.1016/j.jclepro.2019.06.226>
- Li R, Teng JT (2018) Pricing and lot-sizing decisions for perishable goods when demand depends on selling price, reference price, product freshness, and displayed stocks. *Eur J Oper Res* 270:1099–1108. <https://doi.org/10.1016/j.ejor.2018.04.029>
- Li Z, Zhang C (2024) Designing a two-stage model for the resilient agri-food supply chain network under dynamic competition. *Br Food J* 126:662–681. <https://doi.org/10.1108/BFJ-12-2022-1135>
- Li S, Han W, Liu L (2022) Robust optimization of the hub location problem for fresh agricultural products with uncertain demand. *IEEE Access* 10:41902–41913. <https://doi.org/10.1109/ACCESS.2022.3168006>
- Lu TH, Lai CH, Chiang TL (2012) Reducing regional inequality in mortality from road traffic injuries through enforcement of the mandatory motorcycle helmet law in Taiwan. *Inj Prev* 18:150–157. <https://doi.org/10.1136/injuryprev-2011-040135>
- Lu J, Zhang J, Lu F, Tang W (2020) Optimal pricing on an age-specific inventory system for perishable items. *Oper Res* 20:605–625. <https://doi.org/10.1007/s12351-017-0366-x>
- Mallidis I, Sariannidis N, Vlachos D et al (2022) Optimal inventory control policies for avoiding food waste. *Oper Res* 22:685–701. <https://doi.org/10.1007/s12351-020-00554-w>
- Manne AS (1958) Programming of economic lot sizes. *Manage Sci* 4:115–135. <https://doi.org/10.1287/mnsc.4.2.115>
- Melega GM, de Araujo SA, Morabito R (2020) Mathematical model and solution approaches for integrated lot-sizing, scheduling and cutting stock problems. *Ann Oper Res* 295:695–736. <https://doi.org/10.1007/s10479-020-03764-9>
- Mercier S, Villeneuve S, Mondor M, Uysal I (2017) Time-temperature management along the food cold chain: a review of recent developments. *Compr Rev Food Sci Food Saf* 16:647–667. <https://doi.org/10.1111/1541-4337.12269>
- Minagricultura (2019a) Plan Integral de Desarrollo Agropecuario y Rural con enfoque territorial

- Minagricultura (2019b) Agronet. <https://www.agronet.gov.co/estadistica/Paginas/home.aspx?cod=4#collapse4>. Accessed 30 Jan 2020
- Ministerio de Agricultura y Desarrollo Rural, Universidad Sergio Arboleda (2018) Evaluaciones Agropecuarias del Ministerio de Agricultura y Desarrollo Rural
- Miranda PA, Garrido RA (2004) Incorporating inventory control decisions into a strategic distribution network design model with stochastic demand. *Transp Res Part E Logist Transp Rev* 40:183–207. <https://doi.org/10.1016/j.tre.2003.08.006>
- Miranda PA, Garrido RA, Ceroni JA (2009) e-Work based collaborative optimization approach for strategic logistic network design problem. *Comput Ind Eng* 57:3–13. <https://doi.org/10.1016/j.cie.2008.12.007>
- Mirzaei S, Seifi A (2015) Considering lost sale in inventory routing problems for perishable goods. *Comput Ind Eng* 87:213–227. <https://doi.org/10.1016/j.cie.2015.05.010>
- Mogale DG, Cheikhrouhou N, Tiwari MK (2019) Modelling of sustainable food grain supply chain distribution system: a bi-objective approach. *Int J Prod Res*. <https://doi.org/10.1080/00207543.2019.1669840>
- Musavi MM, Bozorgi-Amiri A (2017) A multi-objective sustainable hub location-scheduling problem for perishable food supply chain. *Comput Ind Eng* 113:766–778. <https://doi.org/10.1016/j.cie.2017.07.039>
- Departamento Nacional de Planeación (2016) Pérdida y Desperdicio de alimentos en Colombia
- Nasrollahi S, Hosseini-Nasab H, Fakhrazad MB, Honarvar M (2023) A multi-stage stochastic model for designing a linked cross-docking distribution network with heterogeneous trucks. *Oper Res* 23:15. <https://doi.org/10.1007/s12351-023-00762-0>
- Orozco-Fontalvo M, Cantillo V, Miranda PA (2019) A stochastic, multi-commodity multi-period inventory-location problem: modeling and solving an industrial application. In: Paternina-Arboleda C, Voß S (eds) *Computational logistics*. Springer, Cham, pp 317–331
- Otrodi F, Ghasemy Yaghin R, Torabi SA (2019) Joint pricing and lot-sizing for a perishable item under two-level trade credit with multiple demand classes. *Comput Ind Eng* 127:761–777. <https://doi.org/10.1016/j.cie.2018.11.015>
- Pahl J, Voß S (2014) Integrating deterioration and lifetime constraints in production and supply chain planning: A survey. *Eur J Oper Res* 238:654–674. <https://doi.org/10.1016/j.ejor.2014.01.060>
- Pal S, Mahapatra GS, Samanta GP (2014) An EPQ model of ramp type demand with Weibull deterioration under inflation and finite horizon in crisp and fuzzy environment. *Int J Prod Econ* 156:159–166. <https://doi.org/10.1016/j.ijpe.2014.05.007>
- Peleg M, Engel R, Gonzalez-Martinez C, Corradini MG (2002) Non-Arrhenius and non-WLF kinetics in food systems. *J Sci Food Agric* 82:1346–1355. <https://doi.org/10.1002/jsfa.1175>
- Priefer C, Jörissen J, Bräutigam KR (2016) Food waste prevention in Europe - a cause-driven approach to identify the most relevant leverage points for action. *Resour Conserv Recycl* 109:155–165. <https://doi.org/10.1016/j.resconrec.2016.03.004>
- Qin Y, Wang J, Wei C (2014) Joint pricing and inventory control for fresh produce and foods with quality and physical quantity deteriorating simultaneously. *Int J Prod Econ* 152:42–48. <https://doi.org/10.1016/j.ijpe.2014.01.005>
- Rapolu CN, Kandpal DH (2020) Joint pricing, advertisement, preservation technology investment and inventory policies for non-instantaneous deteriorating items under trade credit. *Opsearch* 57:274–300. <https://doi.org/10.1007/s12597-019-00427-7>
- Rekabi S, Ghodrtnama A, Azaron A (2021) Designing pharmaceutical supply chain networks with perishable items considering congestion. *Oper Res*. <https://doi.org/10.1007/s12351-021-00674-x>
- Rohmer SUK, Claassen GDH, Laporte G (2019) A two-echelon inventory routing problem for perishable products. *Comput Oper Res* 107:156–172. <https://doi.org/10.1016/j.cor.2019.03.015>
- Romeijn HE, Sharkey TC, Shen ZM, Zhang J (2010) Integrating facility location and production planning decisions. *Networks* 55:78–89. <https://doi.org/10.1002/net.20315>
- Rong A, Akkerman R, Grunow M (2011) An optimization approach for managing fresh food quality throughout the supply chain. *Int J Prod Econ* 131:421–429. <https://doi.org/10.1016/j.ijpe.2009.11.026>
- Salin V (1998) Information technology in agri-food supply chains. *Int Food Agribus Manag Rev* 1:329–334. [https://doi.org/10.1016/S1096-7508\(99\)80003-2](https://doi.org/10.1016/S1096-7508(99)80003-2)
- Samper B, Manjarrez L, Álvarez M, et al. (2005) Encuesta Nacional de la situación nutricional en Colombia ENSIN 2005. *Inst Colomb Bienestar Fam República Colomb* 65–66

- Shafiee M, Zare-mehrjerdi Y, Govindan K (2022) A causality analysis of risks to perishable product supply chain networks during the COVID-19 outbreak era : an extended DEMATEL method under Pythagorean fuzzy environment. *Transp Res Part E* 163:102759. <https://doi.org/10.1016/j.tre.2022.102759>
- Shen Z, Dessouky M, Ordóñez F (2011) Perishable inventory management system with a minimum volume constraint. *J Oper Res Soc* 62:2063–2082. <https://doi.org/10.1057/jors.2010.181>
- Singh Z, Singh RK, Sane VA, Nath P (2013) Mango - postharvest biology and biotechnology. *CRC Crit Rev Plant Sci* 32:217–236. <https://doi.org/10.1080/07352689.2012.743399>
- Skouri K, Konstantaras I, Papachristos S, Ganas I (2009) Inventory models with ramp type demand rate, partial backlogging and Weibull deterioration rate. *Eur J Oper Res* 192:79–92. <https://doi.org/10.1016/j.ejor.2007.09.003>
- van der Vorst JG (2000) Effective food supply chains; generating, modelling and evaluating supply chain scenarios. Wageningen University
- Wagner HM, Whitin TM (1958) Dynamic version of the economic lot size model. *Manage Sci* 5:89–96. <https://doi.org/10.1287/mnsc.5.1.89>
- Wu K (2001) An EOQ inventory model for items with Weibull distribution deterioration, ramp type demand rate and partial backlogging. *Prod Plan Control* 12:787–793. <https://doi.org/10.1080/0953728011005181>
- Wu W, Cronjé P, Verboven P, Defraeye T (2019) Unveiling how ventilated packaging design and cold chain scenarios affect the cooling kinetics and fruit quality for each single citrus fruit in an entire pallet. *Food Packag Shelf Life* 21:100369. <https://doi.org/10.1016/j.fpsl.2019.100369>
- Yu M, Nagurny A (2013) Competitive food supply chain networks with application to fresh produce. *Eur J Oper Res* 224:273–282. <https://doi.org/10.1016/j.ejor.2012.07.033>
- Zahiri B, Julia P, Tavakkoli-Moghaddam R (2018) Design of a pharmaceutical supply chain network under uncertainty considering perishability and substitutability of products. *Inf Sci (Ny)* 423:257–283. <https://doi.org/10.1016/j.ins.2017.09.046>

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